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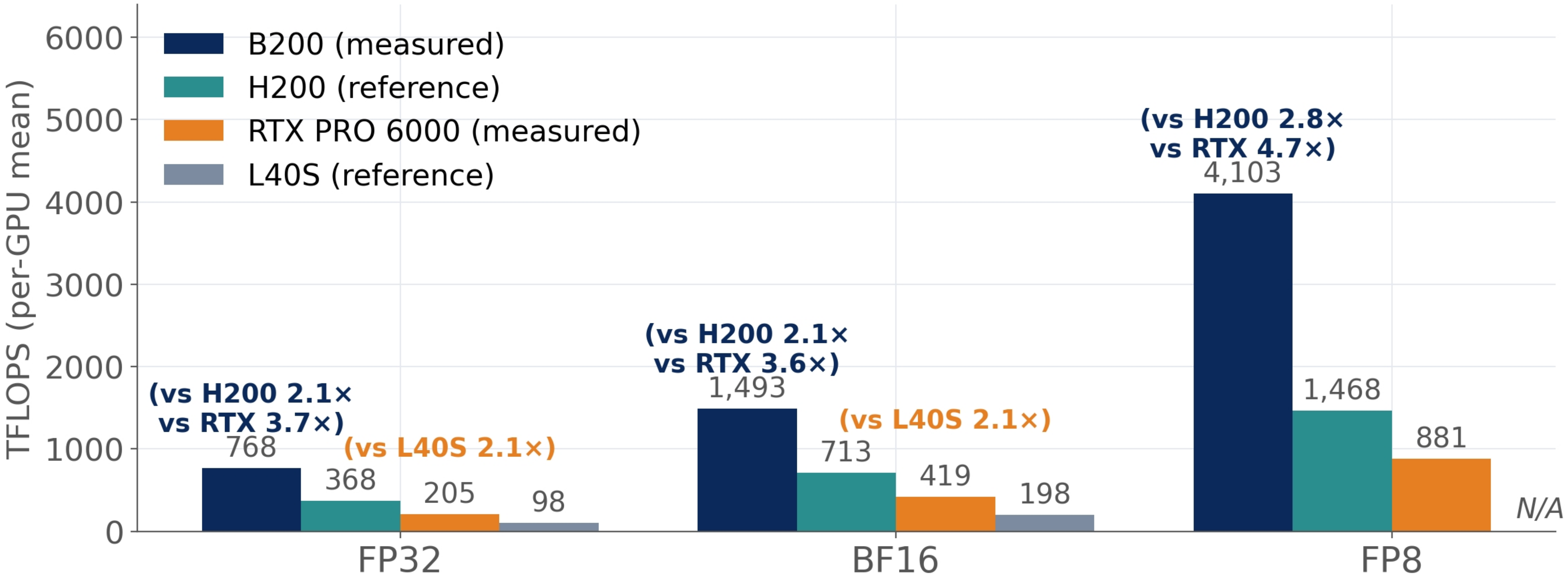
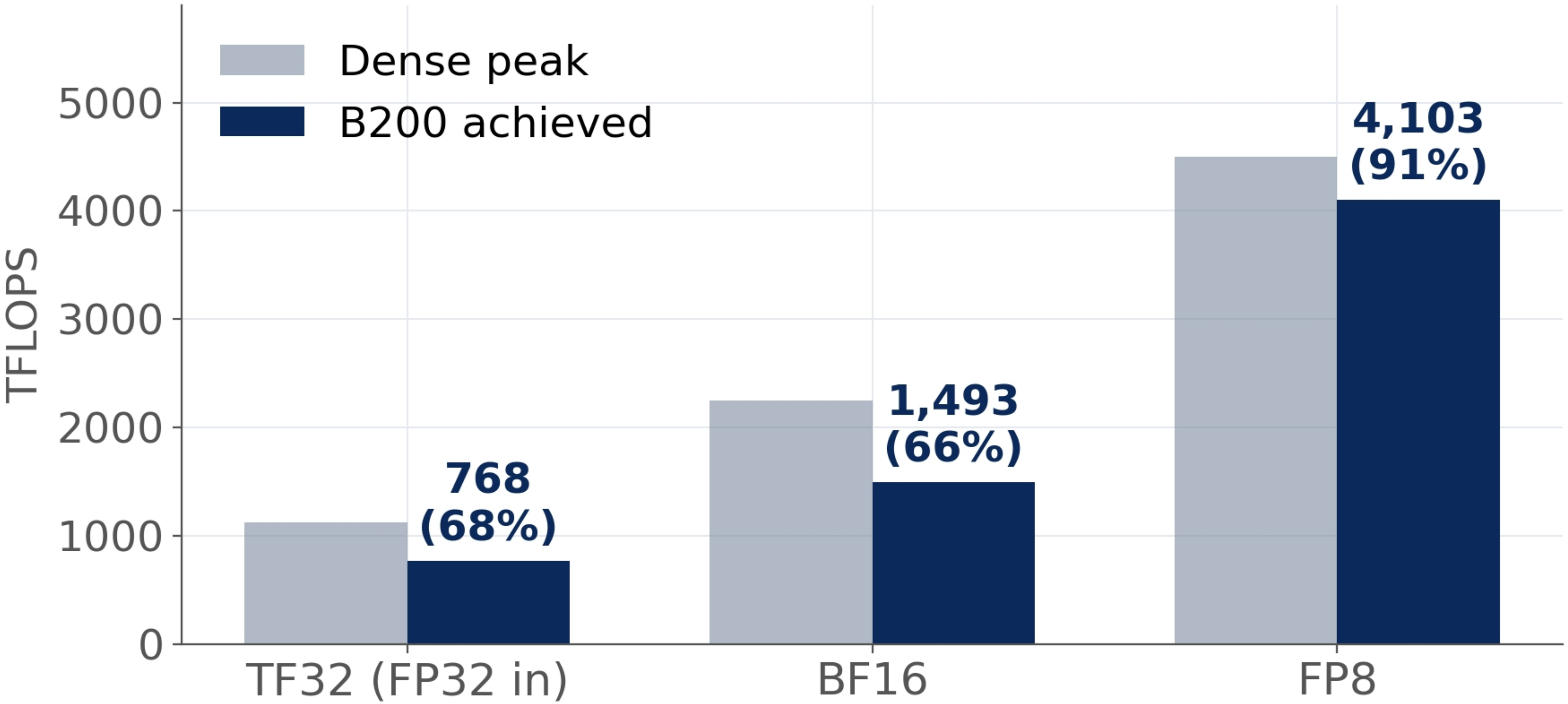
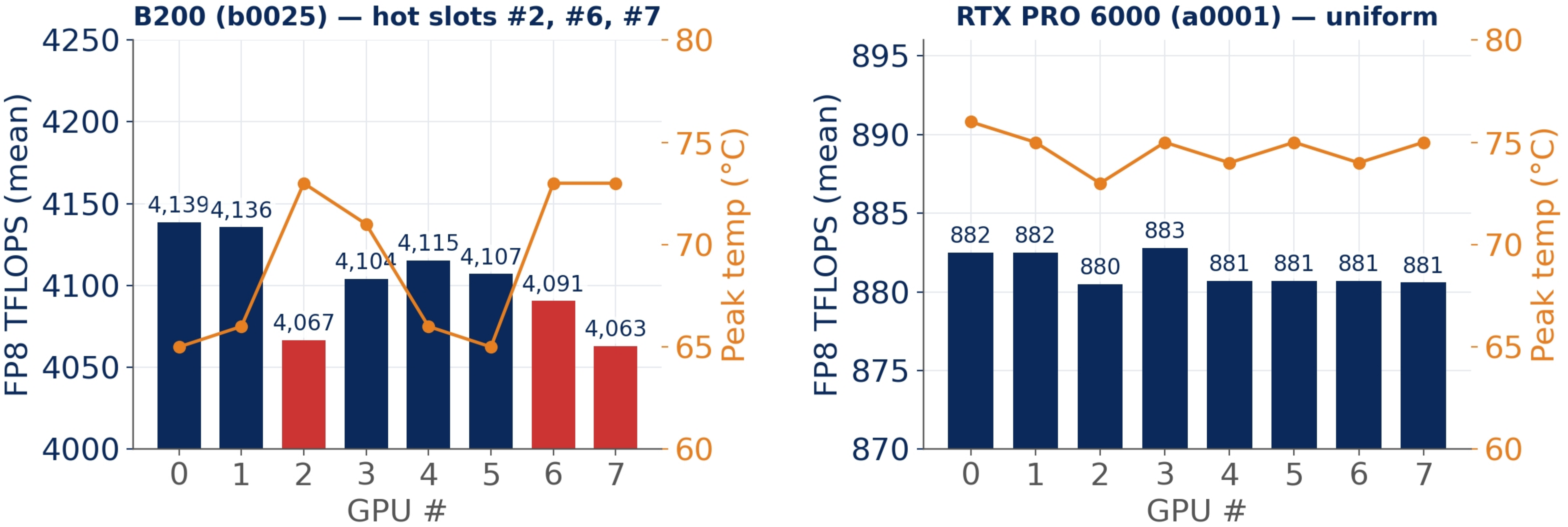
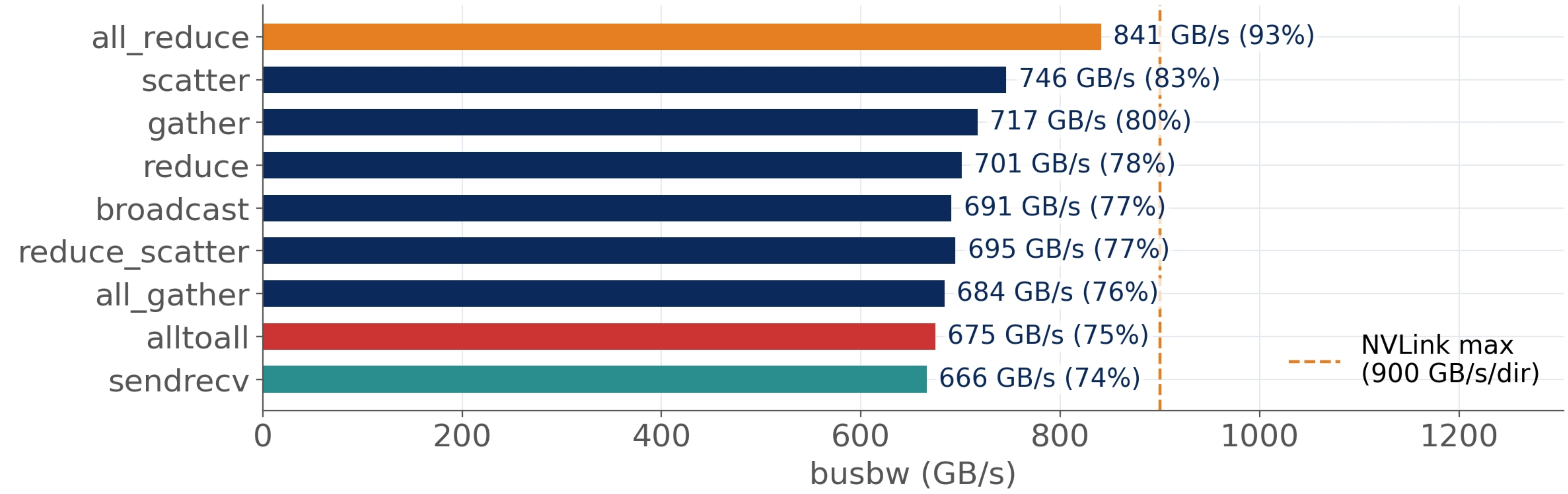


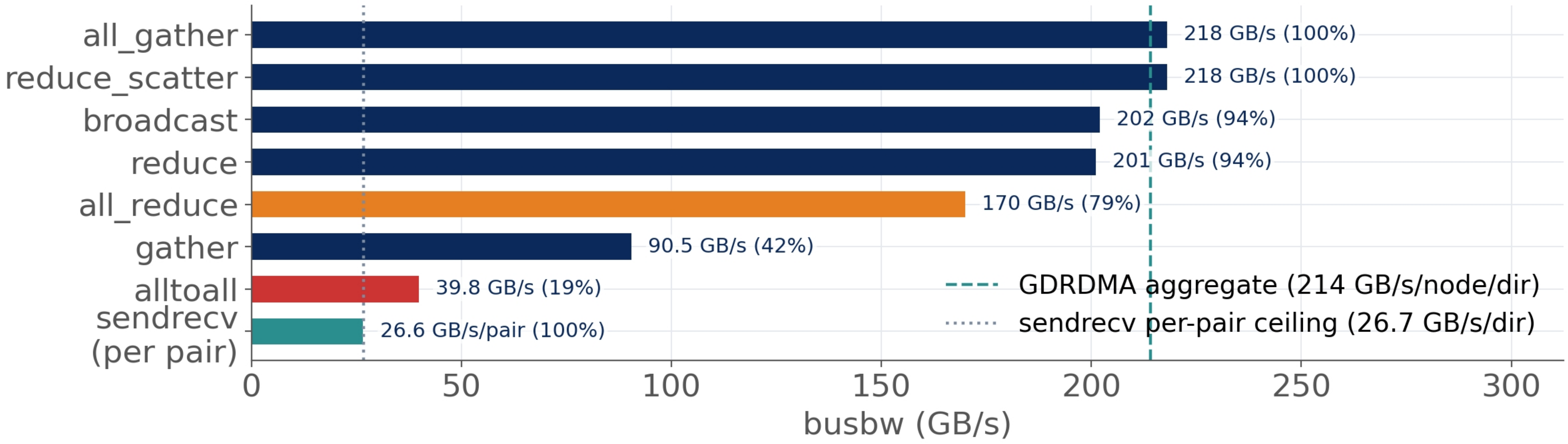
B200 (measured)
H200 (reference)
RTX PRO 6000 (measured)
L40S (reference)
TFLOPS (per-GPU mean)
6000
5000
4000
3000
2000
1000
0
(vs H200 2.1×
vs RTX 3.7×)
768
368
205
(vs L40S 2.1×)
98
(vs H200 2.1×
vs RTX 3.6×)
1,493
713
419
(vs L40S 2.1×)
198
(vs H200 2.8×
vs RTX 4.7×)
4,103
1,468
881
N/A
FP32
BF16
FP8

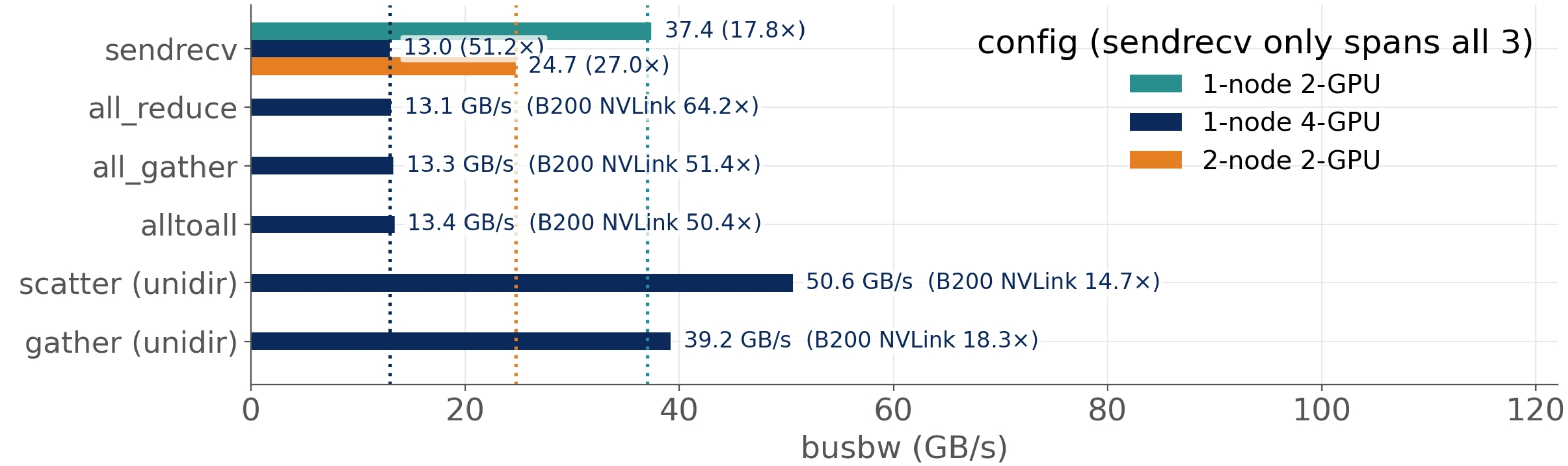

Dense peak
B200 achieved
TFLOPS
5000
4000
3000
2000
1000
0
768 (68%)
1,493 (66%)
4,103 (91%)
TF32 (FP32 in)
BF16
FP8



B200 (b0025) — hot slots #2, #6, #7
FP8 TFLOPS (mean)
Peak temp (°C)
GPU #
4,139
4,136
4,067
4,104
4,115
4,107
4,091
4,063
4250
4200
4150
4100
4050
4000
80
75
70
65
60
0 1 2 3 4 5 6 7
RTX PRO 6000 (a0001) — uniform
FP8 TFLOPS (mean)
Peak temp (°C)
GPU #
882
882
880
883
881
881
881
881
895
890
885
880
875
870
80
75
70
65
60
0 1 2 3 4 5 6 7


all_reduce
scatter
gather
reduce
broadcast
reduce_scatter
all_gather
alltoall
sendrecv
841 GB/s (93%)
746 GB/s (83%)
717 GB/s (80%)
701 GB/s (78%)
691 GB/s (77%)
695 GB/s (77%)
684 GB/s (76%)
675 GB/s (75%)
666 GB/s (74%)
NVLink max
(900 GB/s/dir)
0
200
400
600
800
1000
1200
busbw (GB/s)


all_gather
reduce_scatter
broadcast
reduce
all_reduce
gather
alltoall
sendrecv
(per pair)
218 GB/s (100%)
218 GB/s (100%)
202 GB/s (94%)
201 GB/s (94%)
170 GB/s (79%)
90.5 GB/s (42%)
39.8 GB/s (19%)
26.6 GB/s/pair (100%)
GDRDMA aggregate (214 GB/s/node/dir)
sendrecv per-pair ceiling (26.7 GB/s/dir)
0
50
100
150
200
250
300
busbw (GB/s)



config (sendrecv only spans all 3)
1-node 2-GPU
1-node 4-GPU
2-node 2-GPU
sendrecv
all_reduce
all_gather
alltoall
scatter (unidir)
gather (unidir)
37.4 (17.8×)
13.0 (51.2×)
24.7 (27.0×)
13.1 GB/s (B200 NVLink 64.2×)
13.3 GB/s (B200 NVLink 51.4×)
13.4 GB/s (B200 NVLink 50.4×)
50.6 GB/s (B200 NVLink 14.7×)
39.2 GB/s (B200 NVLink 18.3×)
0
20
40
60
80
100
120
busbw (GB/s)

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[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

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